Zhuoyan Luo

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Education

Tsinghua University Shenzhen International Graduate School 2023 – 2026

Master student in Artificial Intelligence

Southeast University, Computer Science and Engineering Department 2019 – 2023

B.S. in Artificial Intelligence
GPA: 4.08/4.8 Ranking: 5/91

Experience

ATH Department Alibaba Group 2026.07 – Present

Researcher Leader: Xinlong Wang

Brief introduction: Core Contributor in designing the new era of Video Generation Model.

ATH Department Alibaba Group 2026.01 – 2026.07

Research Intern Adviser: Xinlong Wang

Brief introduction: Core Contributor in designing the Video Generation Model.
Project: **Happy Horse 1.0** (LLM Arena Rank2)

BAAI Vision 2025.02 – 2026.01

Research Intern in EMU Team Adviser: Xinlong Wang

Brief introduction: Core contributor in designing a new general visual tokenizer for EMU model.
Project Repository: **Emu3.5** (Github Stars 1.6k+)

Tencent ARC Laboratory 2024.01 – 2025.02

Research Intern Adviser: Yixiao Ge

Brief introduction: Core contributor in the Visual Tokenizer training and AutoRegressive Image Generation for the Multi-Modal Large Language Model.
Project Repository: **SEED-Voken** (Github Stars 1k+)

Selected Publications

- Yinghao Wu*, **Zhuoyan Luo***, Yiyao Yu, Zhaojian Yu, Yujiu Yang and Xiao-Ping Zhang. VEN-VL: A Visual Ensemble MoE Framework for Effective and Efficient Multi-Modal Understanding. In ***EMNLP***, 2026.
- Emu3.5 Team (The Core Contributor). Emu3.5: Native Multimodal Models are World Learners. ***Technical Report*** In Arxiv, 2025.
- Yong Liu*, **Zhuoyan Luo***, Yicheng Xiao*, Yitong Wang, Shuyan Li, Xiu Li, Yujiu Yang and Yansong Tang. Semantic-Assisted Object Clustering for Multi-Modal Referring Video Segmentation. In ***TPAMI***, 2025.
- Wenxuan Wang, Fan Zhang, Yufeng Cui, Haiwen Diao, **Zhuoyan Luo**, Huchuan Lu, Jing Liu, Xinlong Wang. End-to-End Vision Tokenizer Tuning. In ***NeurIPS***, 2025.
- Fengyuan Shi*, **Zhuoyan Luo***, Yixiao Ge, Yujiu Yang, Ying Shan, Limin Wang. Scalable Image Tokenization with Index Backpropagation Quantization. In ***ICCV***, 2025.
- **Zhuoyan Luo**, Fengyuan Shi, Yixiao Ge, Yujiu Yang, Limin Wang, Ying Shan. OPEN-MAGVIT2: An Open-source Project Toward Democratizing Auto-Regressive Visual Generation. In ***Arxiv***, 2024.
- **Zhuoyan Luo**, Yinghao Wu, Yong Liu, Yicheng Xiao, Xiao-Ping Zhang, Yujiu Yang. CoHD: A Counting-Aware Hierarchical Decoding Framework for Generalized Referring Expression Segmentation. In ***ICCV***, 2025.

- Yicheng Xiao*, **Zhuoyan Luo***, Yong Liu, Yue Ma, Hengwei Bian, Yatai Ji, Yujiu Yang, Xiu Li. Bridging the Gap: A Unified Video Comprehension Framework for Moment Retrieval and Highlight Detection. In ***CVPR***, 2024.
- **Zhuoyan Luo**, Yicheng Xiao, Yong Liu, Shuyan Li, Yitong Wang, Yansong Tang, Xiu Li, Yujiu Yang. SOC: Semantic-Assisted Object Cluster for Referring Video Object Segmentation. In ***NeurIPS***, 2023.
- **Zhuoyan Luo**, Yicheng Xiao, Yong Liu, Yitong Wang, Yansong Tang, Xiu Li, Yujiu Yang. 1st Place Solution for Referring Video Object Segmentation. In ***ICCV Workshop***, 2023.
- **Zhuoyan Luo**, Yinghao Wu, Shipeng Zhu, Qi Jing, Zihang Xu, Hui Xue. 2nd Place Solution for Multi-line Handwritten Mathematical Expression Recognition Competition. In ***ICDAR Workshop***, 2023.
- Qipeng Zhu*,**Zhuoyan Luo***, Shipeng Zhu*, Qi Jing, Zihang Xu, Hui Xue. FATE: A Three-Stage Method for Arithmetical Exercise Correction. In the Journal of ***Neural Computing and Applications***, 2023.
- **Zhuoyan Luo**, Hui Xue. Deep Neural Network Based Mathematical Exercises Recognition and Automatic Correction Algorithm. China, 202210783495.4.

AWARDS

Outstanding Graduate Student Award in Tsinghua University	July 2026
1st Prize Award on ICCV 2023 The 5th Large-scale Video Object Segmentation Challenge	Oct. 2023
2nd Prize Award on ICDAR 2023 Multi-line Handwritten Mathematical Expression Recognition	Mar. 2023
2nd Prize Award on iFLYTEK A.I. Handwritten Mathematical Expression Recognition	Nov. 2022
***Bronze* Prize** Award on XIB Competition	May 2024
National Scholarship	2021-2022
Principal Scholarship	2022-2023

PERSONAL

- Research Interest: Multi-Modal Understanding & Generation
- Great Teamwork and Compressive Ability